Motivations, Barriers and ActiveSG Gym Use by Age and Sex, 2022–2025

NMC5344 Group Project Report

Group 4
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Word count: 1,064 (excluding R code, output, tables and references)

```
library(tidyverse)   # readr, dplyr, tidyr, stringr, ggplot2
library(knitr)       # kable() tables
library(broom)       # tidy() and glance() model summaries

YEARS      <- 2022:2025
AGE_LEVELS <- c("13-19", "20-39", "40-59", "60+")
```

Introduction

Many people know exercise is important, yet not everyone acts on it. ActiveSG data show that from 2022 to 2025, gym visits by women grew much faster than men's, and those by adults aged 60+ grew fastest.

Research question: How do motivations and barriers to exercise relate to changing ActiveSG gym attendance across age and sex between 2022 and 2025?

Following the instructor's feedback on our proposal, we analyse groups, divide visits by population, use four survey indices and check for Simpson's paradox.

1. Datasets and data preparation

1a. Observations and variables

We use three public datasets (Department of Statistics Singapore, n.d.; Sport Singapore, n.d.-a, n.d.-b, 2026).

```
mot_raw <- read_csv("MotivationstoParticipateinSportExercise.csv")
bar_raw <- read_csv("BarrierstoParticipationinSportExercise.csv")

# The SingStat file has notes above and below the table; readLines() + read.csv()
# keeps the leading spaces in "Data Series", which mark region, sex and age rows.
gym_lines <- readLines("M920481.csv", warn = FALSE)
header    <- grep("^Data Series", gym_lines)
gym_raw   <- read.csv(text = gym_lines[header:length(gym_lines)],
```

```
                      colClasses = "character", check.names = FALSE,
                      strip.white = FALSE)
names(gym_raw) <- trimws(names(gym_raw))
gym_raw <- gym_raw |>
  filter(!is.na(suppressWarnings(as.numeric(`2025`))))   # keep data rows only

tibble(Dataset      = c("Motivations survey", "Barriers survey", "ActiveSG gym visits"),
       Observations = c(nrow(mot_raw), nrow(bar_raw), nrow(gym_raw)),
       Variables    = c(ncol(mot_raw), ncol(bar_raw), ncol(gym_raw)),
       Years        = c("2019-2025", "2019-2025", "2022-2025")) |>
  kable(booktabs = TRUE, caption = "Size of each dataset")
```

Table 1: Size of each dataset

Dataset	Observations	Variables	Years
Motivations survey	560	5	2019-2025
Barriers survey	555	5	2019-2025
ActiveSG gym visits	83	5	2022-2025

Each survey row gives the % citing one reason in one year, age group and sex (motivations: participants; barriers: non-participants). The gym table has one row per region, sex or age series.

1b. Variable types

```
var_types <- function(data, name) {
  tibble(Dataset = name, Variable = names(data),
         Type = map_chr(data, ~ class(.x)[1]))
}

bind_rows(var_types(mot_raw, "Motivations"),
          var_types(bar_raw, "Barriers"),
          var_types(gym_raw, "Gym visits")) |>
  kable(booktabs = TRUE, caption = "Variable types in the raw datasets")
```

Table 2: Variable types in the raw datasets

Dataset	Variable	Type
Motivations	Year	numeric
Motivations	AgeGroup	character
Motivations	Sex	character
Motivations	Motivations	character
Motivations	Percentage	numeric
Barriers	Year	numeric
Barriers	AgeGroup	character
Barriers	Sex	character
Barriers	Barriers	character
Barriers	Percentage	numeric
Gym visits	Data Series	character
Gym visits	2025	character
Gym visits	2024	character
Gym visits	2023	character

Dataset	Variable	Type
Gym visits	2022	character

Year and Percentage are numeric; age group, sex and reason are categorical. Gym visit counts are read as text and converted below.

1c. Data preparation

Harmonising age groups. The same group appears as "13 to 19", "13 -19" and "13 - 19", so we remove spaces and replace "to" with "-".

```
clean_age <- function(x) str_replace(str_remove_all(x, "\\s"), "to", "-")
```

Survey indices. Four indices, chosen before modelling, combine related items: health motivation (physical or mental health), enjoyment motivation (interest in the sport or meeting people), time barriers (work, family, school) and health barriers (too old or health condition). The surveys publish only each group's top 10 items, so a missing item is coded 0, a lower bound rather than an observed value (table below; Section 4c tests this).

```
mot_items <- c(physical = "better physical health", mental = "better mental health",
               interest = "interest of the sports", social = "meet other people")
bar_items <- c(overtime = "too busy working overtime", family = "take care of family members",
               school = "too busy with schoolwork", studywork = "studying and working",
               twojobs = "more than one job", too_old = "too old for sport",
               health_cnd = "health condition is not suitable")

# For each item, find its rows and place the % in a 32-row grid (NA = not in top 10)
extract_items <- function(df, item_col, patterns) {
  df <- df |>
    rename(item = all_of(item_col)) |>
    filter(Year %in% YEARS) |>
    mutate(AgeGroup = clean_age(AgeGroup), item = str_to_lower(item))
  grid <- expand_grid(Year = YEARS, AgeGroup = AGE_LEVELS, Sex = c("Female", "Male"))
  for (nm in names(patterns)) {
    matched <- df |>
      filter(str_detect(item, fixed(patterns[[nm]]))) |>
      group_by(Year, AgeGroup, Sex) |>
      summarise(!!nm := sum(Percentage), .groups = "drop")
    grid <- left_join(grid, matched, by = c("Year", "AgeGroup", "Sex"))
  }
  grid
}

items <- extract_items(mot_raw, "Motivations", mot_items) |>
  left_join(extract_items(bar_raw, "Barriers", bar_items),
            by = c("Year", "AgeGroup", "Sex"))

item_labels <- c(
  physical = "M: better physical health", mental = "M: better mental health",
  interest = "M: interest in the sport", social = "M: meet other people",
  overtime = "B: busy working overtime", family = "B: care for family",
  school = "B: busy with schoolwork", studywork = "B: studying and working",
  twojobs = "B: more than one job", too_old = "B: too old for sport",
  health_cnd = "B: health condition")
```

```
items |>
  summarise(across(-c(Year, AgeGroup, Sex), ~ sum(is.na(.x)))) |>
  pivot_longer(everything(), names_to = "Item", values_to = "Cells set to 0 (of 32)") |>
  mutate(Item = item_labels[Item]) |>
  kable(booktabs = TRUE, caption = "Survey items missing from a group's top 10")
```

Table 3: Survey items missing from a group's top 10

Item	Cells set to 0 (of 32)
M: better physical health	0
M: better mental health	0
M: interest in the sport	0
M: meet other people	1
B: busy working overtime	6
B: care for family	8
B: busy with schoolwork	19
B: studying and working	23
B: more than one job	19
B: too old for sport	19
B: health condition	5

```
items <- items |> mutate(across(-c(Year, AgeGroup, Sex), ~ replace_na(.x, 0)))

survey <- items |>
  transmute(Year, AgeGroup, Sex,
            mot_health = (physical + mental) / 2,                      # % of participants
            mot_enjoy  = (interest + social) / 2,
            bar_time   = overtime + family + school + studywork + twojobs,  # per 100 non-participants
            bar_health = too_old + health_cnd)
```

Gym visits. The table marks region, sex and age rows only by indentation, so we count leading spaces, carry labels down with `fill()`, and sum over the five regions.

```
gym_long <- gym_raw |>
  rename(series = `Data Series`) |>
  mutate(indent = nchar(series) - nchar(trimws(series, which = "left")),
         label  = trimws(series),
         region = if_else(indent == 0, label, NA_character_),
         sex    = if_else(indent == 2, label, NA_character_)) |>
  fill(region) |>
  group_by(region) |> fill(sex) |> ungroup() |>
  pivot_longer(as.character(YEARS), names_to = "Year", values_to = "visits") |>
  mutate(Year = as.integer(Year), visits = as.numeric(visits))
```

The 2024 gap. About 1.47 million visits in July–December 2024 have no region, sex or age. We allocate them by each group's share of recorded 2024 visits; otherwise 2024 would show a false drop.

```
year_totals <- gym_long |>
  filter(indent == 0) |>
  group_by(Year) |>
  summarise(grand_total = sum(visits))

age_sex <- gym_long |>
```

```
  filter(indent == 4, sex %in% c("Female", "Male"),
         label %in% c("13 - 19", "20 - 39", "40 - 59", "60+")) |>
  mutate(AgeGroup = clean_age(label)) |>
  group_by(Year, AgeGroup, Sex = sex) |>
  summarise(visits_recorded = sum(visits), .groups = "drop")

allocation <- age_sex |>
  group_by(Year) |>
  summarise(recorded = sum(visits_recorded)) |>
  left_join(year_totals, by = "Year") |>
  mutate(scale_factor = grand_total / recorded)

allocation |>
  mutate(Year = as.character(Year)) |>
  kable(booktabs = TRUE, digits = 3, format.args = list(big.mark = ","),
        caption = "Allocation of visits with no age or sex recorded")
```

Table 4: Allocation of visits with no age or sex recorded

Year	recorded	grand_total	scale_factor
2022	3,031,775	3,053,985	1.007
2023	4,010,212	4,020,748	1.003
2024	2,600,702	4,076,128	1.567
2025	5,146,127	5,147,725	1.000

Visits per resident. Total visits largely reflect group size, so we divide by each group's resident population (SingStat table M810011, end June). The 13–19 population is two-fifths of the 10–14 band (assuming ages are evenly spread) plus the 15–19 band.

```
population <- read_csv("population.csv")

merged <- age_sex |>
  left_join(select(allocation, Year, scale_factor), by = "Year") |>
  mutate(visits = visits_recorded * scale_factor) |>
  inner_join(survey, by = c("Year", "AgeGroup", "Sex")) |>
  left_join(population, by = c("Year", "AgeGroup", "Sex")) |>
  mutate(year  = Year,
         age   = factor(AgeGroup, levels = AGE_LEVELS),
         sex   = factor(Sex, levels = c("Female", "Male")),
         group = paste(AgeGroup, Sex),
         visits_per_resident = visits / Population) |>
  select(year, age, sex, group, visits, population = Population,
         visits_per_resident, mot_health, mot_enjoy, bar_time, bar_health)

stopifnot(nrow(merged) == 32, !anyNA(merged))
glimpse(merged)
```

```
## Rows: 32
## Columns: 11
## $ year                <dbl> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 20~
## $ age                 <fct> 13-19, 13-19, 20-39, 20-39, 40-59, 40-59, 60+, 60+~
## $ sex                 <fct> Female, Male, Female, Male, Female, Male, Female, ~
## $ group               <chr> "13-19 Female", "13-19 Male", "20-39 Female", "20-~
```

```
## $ visits              <dbl> 90182.85, 657260.91, 345782.69, 1109456.48, 150857~
## $ population          <dbl> 140953, 146747, 571507, 550012, 615293, 581714, 51~
## $ visits_per_resident <dbl> 0.6398080, 4.4788712, 0.6050367, 2.0171496, 0.2451~
## $ mot_health          <dbl> 52.5, 50.0, 59.5, 55.0, 62.0, 62.0, 56.0, 57.5, 48~
## $ mot_enjoy           <dbl> 29.0, 40.5, 14.5, 26.0, 9.5, 14.5, 9.0, 8.0, 35.5,~
## $ bar_time            <dbl> 42, 52, 57, 58, 42, 47, 14, 13, 53, 52, 63, 71, 47~
## $ bar_health          <dbl> 0, 6, 7, 0, 2, 6, 44, 45, 0, 0, 5, 0, 7, 16, 53, 5~
```

The final dataset has 32 rows (4 years × 4 age groups × 2 sexes) and 11 variables: age group and sex are categorical; all others are numeric.

2. Summary statistics

```
stats5 <- function(data, var) {
  data |> summarise(n = n(), Mean = mean({{ var }}), Median = median({{ var }}),
                    SD = sd({{ var }}), Min = min({{ var }}), Max = max({{ var }}))
}
tab <- function(x, cap, d = 2) kable(x, booktabs = TRUE, digits = d,
                                     format.args = list(big.mark = ","), caption = cap)
```

```
bind_rows(merged |> stats5(visits) |> mutate(Year = "All years"),
          merged |> group_by(Year = year) |> stats5(visits) |> mutate(Year = as.character(Year))) |>
  relocate(Year) |>
  tab("Annual gym visits per age-sex group", d = 0)
```

Table 5: Annual gym visits per age-sex group

Year	n	Mean	Median	SD	Min	Max
All years	32	509,331	379,693	423,403	75,441	1,728,239
2022	8	381,748	287,272	351,023	75,441	1,109,456
2023	8	502,594	382,451	439,760	119,232	1,390,386
2024	8	509,516	422,547	429,635	127,744	1,432,161
2025	8	643,466	569,574	503,487	154,355	1,728,239

```
bind_rows(merged |> group_by(Group = sex) |> stats5(visits_per_resident),
          merged |> group_by(Group = age) |> stats5(visits_per_resident)) |>
  mutate(Group = as.character(Group)) |>
  tab("Gym visits per resident by sex and age group")
```

Table 6: Gym visits per resident by sex and age group

Group	n	Mean	Median	SD	Min	Max
Female	16	0.59	0.56	0.32	0.15	1.11
Male	16	2.40	1.59	2.00	0.50	6.11
13-19	8	3.16	2.77	2.48	0.64	6.11
20-39	8	1.70	1.56	0.98	0.61	3.14
40-59	8	0.63	0.60	0.32	0.25	1.15
60+	8	0.49	0.45	0.29	0.15	0.99

Visits are right-skewed (mean 509,331, median 379,693): a few large groups, especially men aged 20–39, pull the mean up. Per resident, men made about four times as many visits as women (2.40 vs 0.59 a year), and use falls steadily with age, from 3.16 visits per teenager to 0.49 among those aged 60+.

```
items |>
  pivot_longer(-c(Year, AgeGroup, Sex), names_to = "Item", values_to = "pct") |>
  mutate(Item = item_labels[Item]) |>
  group_by(Item) |> stats5(pct) |> arrange(desc(Mean)) |>
  tab("Survey items (% of respondents)", d = 1)
```

Table 7: Survey items (% of respondents)

Item	n	Mean	Median	SD	Min	Max
M: better physical health	32	79.1	79.0	9.3	62	93
M: interest in the sport	32	33.7	28.5	19.5	10	72
M: better mental health	32	31.9	31.0	5.6	21	44
B: busy working overtime	32	19.5	18.0	16.7	0	56
B: care for family	32	16.0	12.5	14.9	0	48
B: busy with schoolwork	32	11.8	0.0	20.4	0	61
B: health condition	32	11.2	7.0	10.9	0	34
M: meet other people	32	9.9	8.0	5.1	0	23
B: too old for sport	32	8.8	0.0	14.1	0	39
B: studying and working	32	3.0	0.0	5.4	0	17
B: more than one job	32	2.0	0.0	2.7	0	9

```
merged |>
  pivot_longer(mot_health:bar_health, names_to = "Index", values_to = "value") |>
  mutate(Index = recode(Index, mot_health = "Health motivation", mot_enjoy = "Enjoyment motivation",
                        bar_time = "Time barrier", bar_health = "Health barrier")) |>
  group_by(Index, `Age group` = age) |> stats5(value) |>
  tab("Motivation and barrier indices by age group", d = 1)
```

Table 8: Motivation and barrier indices by age group

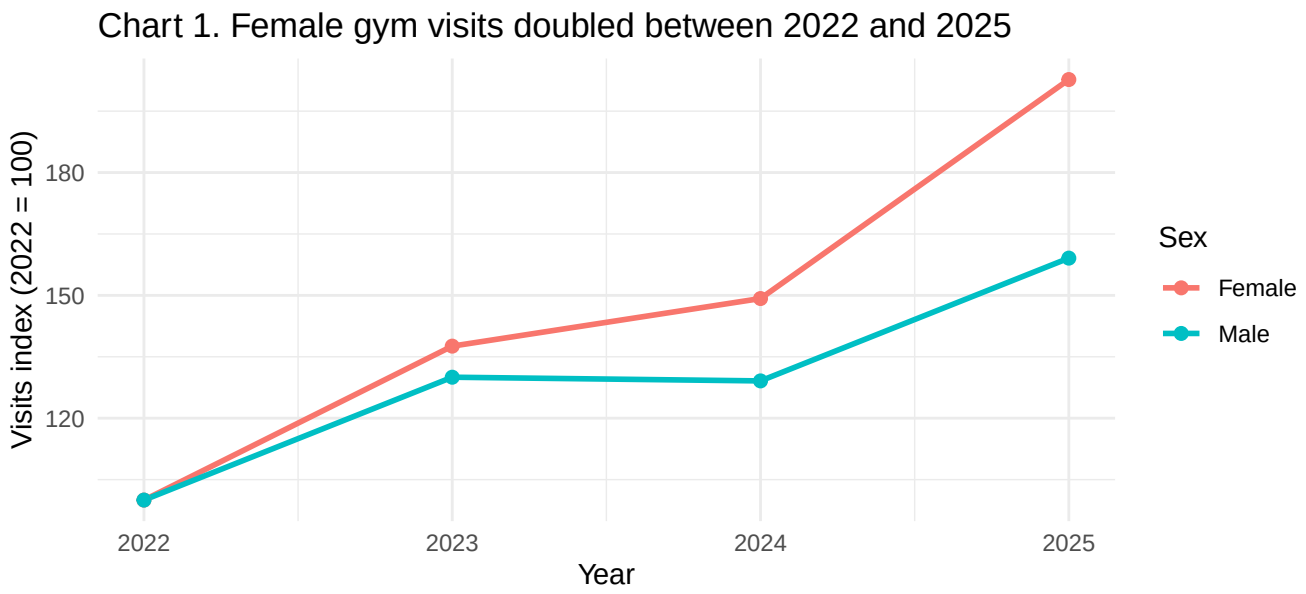
Index	Age group	n	Mean	Median	SD	Min	Max
Enjoyment motivation	13-19	8	38.2	38.2	5.7	29.0	46.0
Enjoyment motivation	20-39	8	24.0	23.8	7.1	14.5	32.0
Enjoyment motivation	40-59	8	14.8	15.2	4.3	9.5	21.5
Enjoyment motivation	60+	8	10.1	9.5	3.6	5.0	17.0
Health barrier	13-19	8	4.5	5.5	4.1	0.0	11.0
Health barrier	20-39	8	4.5	5.0	3.0	0.0	8.0
Health barrier	40-59	8	11.4	10.0	6.8	2.0	20.0
Health barrier	60+	8	59.5	60.5	11.4	44.0	73.0
Health motivation	13-19	8	49.9	49.8	2.0	47.0	52.5
Health motivation	20-39	8	53.8	53.2	5.7	47.0	60.5
Health motivation	40-59	8	59.8	61.8	4.1	52.5	65.0
Health motivation	60+	8	58.6	58.5	1.6	56.0	61.0
Time barrier	13-19	8	58.6	54.5	12.2	42.0	77.0
Time barrier	20-39	8	75.0	71.0	16.0	57.0	94.0
Time barrier	40-59	8	58.6	59.5	12.7	42.0	77.0
Time barrier	60+	8	16.9	15.5	3.8	13.0	24.0

Better physical health is the most common motivation (mean 79%); overtime and family care are the most common

barriers. Age, not sex, divides the indices: enjoyment motivation falls from 38 (teenagers) to 10 (60+), while health barriers rise from under 12 to 60.

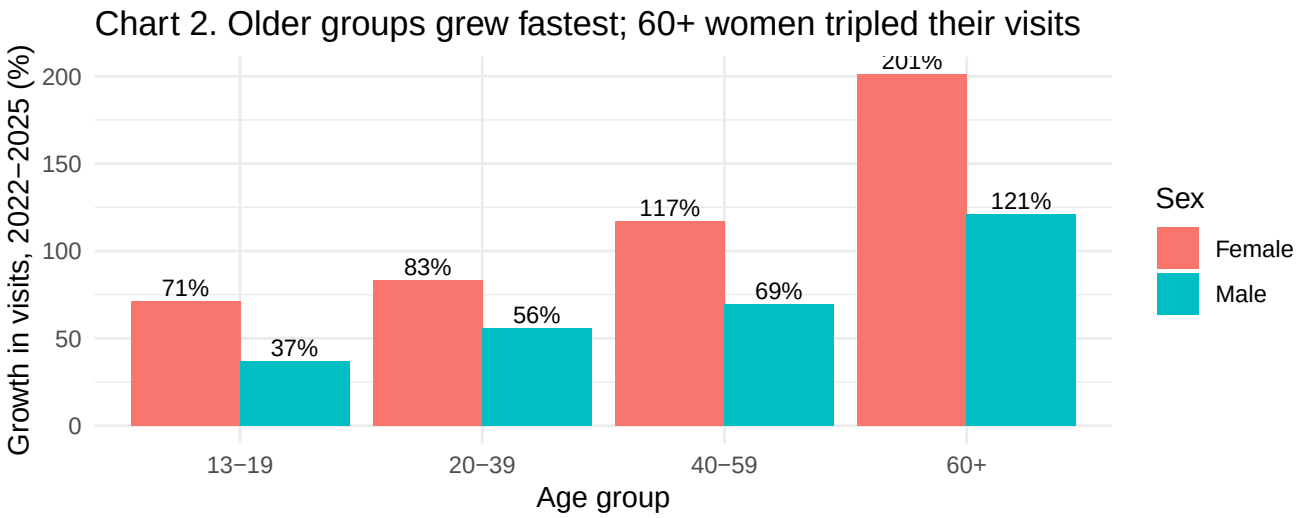
3. Visualisations

```
merged |>
  group_by(year, sex) |> summarise(visits = sum(visits), .groups = "drop") |>
  group_by(sex) |> mutate(index = visits / visits[year == 2022] * 100) |>
  ggplot(aes(year, index, colour = sex)) +
  geom_line(linewidth = 1) + geom_point(size = 2) +
  labs(title = "Chart 1. Female gym visits doubled between 2022 and 2025",
       x = "Year", y = "Visits index (2022 = 100)", colour = "Sex") +
  theme_minimal()
```



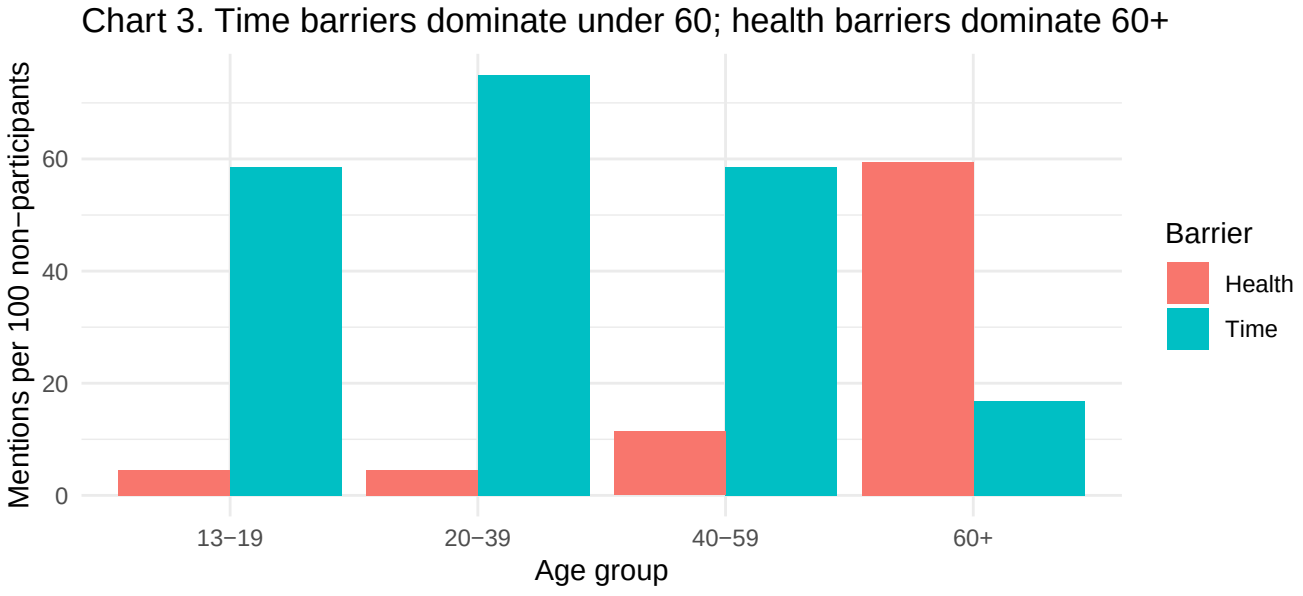
Female visits grew much faster than male visits (+103% vs +59%), and the growth gap widened after 2023, when male visits plateaued.

```
merged |>
  filter(year %in% c(2022, 2025)) |>
  select(year, age, sex, visits) |>
  pivot_wider(names_from = year, values_from = visits, names_prefix = "y") |>
  mutate(growth = (y2025 / y2022 - 1) * 100) |>
  ggplot(aes(age, growth, fill = sex)) +
  geom_col(position = "dodge") +
  geom_text(aes(label = paste0(round(growth), "%")),
            position = position_dodge(width = 0.9), vjust = -0.3, size = 3) +
  labs(title = "Chart 2. Older groups grew fastest; 60+ women tripled their visits",
       x = "Age group", y = "Growth in visits, 2022-2025 (%)", fill = "Sex") +
  theme_minimal()
```



Growth rises with age for both sexes, from 37% (men) and 71% (women) among teenagers to 121% and 201% among those aged 60+.

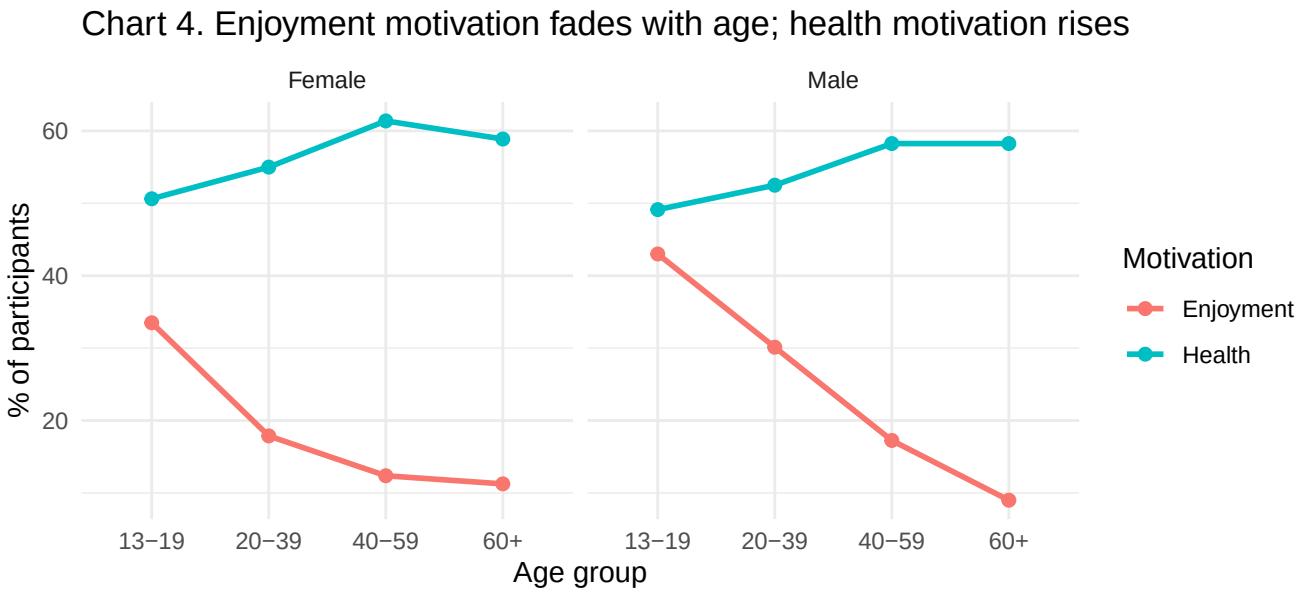
```
merged |>
  group_by(age) |>
  summarise(Time = mean(bar_time), Health = mean(bar_health)) |>
  pivot_longer(-age, names_to = "barrier", values_to = "value") |>
  ggplot(aes(age, value, fill = barrier)) +
  geom_col(position = "dodge") +
  labs(title = "Chart 3. Time barriers dominate under 60; health barriers dominate 60+",
       x = "Age group", y = "Mentions per 100 non-participants", fill = "Barrier") +
  theme_minimal()
```



Health, not time, is the main barrier after 60. Time barriers peak at about 75 mentions per 100 among 20–39 year olds and fall to 17 at 60+.

```
merged |>
  group_by(age, sex) |>
  summarise(Health = mean(mot_health), Enjoyment = mean(mot_enjoy), .groups = "drop") |>
```

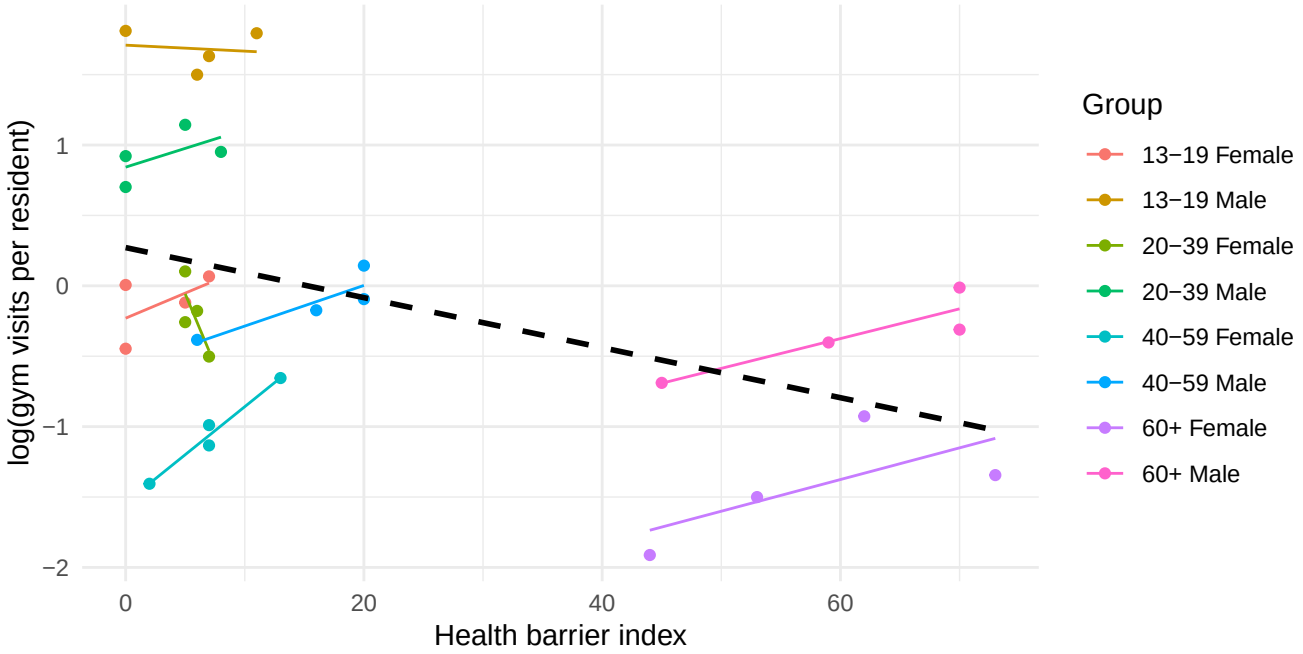
```
  pivot_longer(c(Health, Enjoyment), names_to = "motivation", values_to = "pct") |>
  ggplot(aes(age, pct, colour = motivation, group = motivation)) +
  geom_line(linewidth = 1) + geom_point(size = 2) +
  facet_wrap(~ sex) +
  labs(title = "Chart 4. Enjoyment motivation fades with age; health motivation rises",
       x = "Age group", y = "% of participants", colour = "Motivation") +
  theme_minimal()
```



Motivation shifts from enjoyment to health with age. Enjoyment falls from about 43% to 9% among men and from 34% to 12% among women.

```
ggplot(merged, aes(bar_health, log(visits_per_resident))) +
  geom_point(aes(colour = group)) +
  geom_smooth(aes(colour = group), method = "lm", se = FALSE, linewidth = 0.5) +
  geom_smooth(method = "lm", se = FALSE, colour = "black", linetype = "dashed") +
  labs(title = "Chart 5. The pooled trend (dashed) reverses within groups",
       x = "Health barrier index", y = "log(gym visits per resident)", colour = "Group") +
  theme_minimal()
```

Chart 5. The pooled trend (dashed) reverses within groups



The pooled trend is misleading. Across all groups, higher health barriers go with fewer visits per resident, but within six of eight age-sex groups the link is positive. The pooled line compares older with younger groups: a Simpson's paradox.

4. Models

```
gym <- merged |>
  mutate(year = factor(year),
         male = as.integer(sex == "Male"),
         log_vpr = log(visits_per_resident),
         across(mot_health:bar_health, ~ as.numeric(scale(.x))))    # z-scores
```

The outcome is log(visits per resident), because the rates are right-skewed. Indices are z-scores, so a coefficient b means the visit rate changes by $100 \times (e^b - 1)\%$ per 1 SD increase.

4a. Two models and their motivation

As Chart 5 shows, comparing groups mainly compares young and old people: health motivation's uncontrolled slope is also negative (-0.58) but positive within 7 of 8 groups. Both models therefore control for age-sex group and year, comparing each group only with itself.

Model 1 (control model). Is a group's gym use higher in years when its motivations are stronger or barriers weaker? It adds age-sex group (removing stable differences) and year (removing shared changes such as post-COVID recovery) to the four indices.

```
model_1 <- lm(log_vpr ~ mot_health + mot_enjoy + bar_time + bar_health +
                group + year, data = gym)
summary(model_1)
```

```
##
## Call:
## lm(formula = log_vpr ~ mot_health + mot_enjoy + bar_time + bar_health +
##     group + year, data = gym)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.098841 -0.047604 -0.003346  0.032700  0.155238
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -0.49837    0.10700  -4.658 0.000226 ***
## mot_health         0.06199    0.03219   1.926 0.071049 .
## mot_enjoy          0.42421    0.09457   4.486 0.000325 ***
## bar_time           0.02483    0.05562   0.446 0.660907
## bar_health         0.21029    0.08976   2.343 0.031563 *
## group13-19 Male    1.46979    0.09286  15.828 1.32e-11 ***
## group20-39 Female  0.37644    0.13282   2.834 0.011449 *
## group20-39 Male    1.13623    0.06507  17.461 2.72e-12 ***
## group40-59 Female -0.33370    0.16673  -2.001 0.061569 .
## group40-59 Male    0.37548    0.13204   2.844 0.011222 *
## group60+ Female   -1.03818    0.22064  -4.705 0.000204 ***
## group60+ Male      0.08980    0.23707   0.379 0.709532
## year2023           0.22238    0.05337   4.167 0.000646 ***
## year2024           0.14228    0.09049   1.572 0.134310
## year2025           0.25747    0.09822   2.621 0.017872 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07541 on 17 degrees of freedom
## Multiple R-squared:  0.9967, Adjusted R-squared:  0.994
## F-statistic: 369.3 on 14 and 17 DF,  p-value: < 2.2e-16
```

Model 2 (interaction model). Our question asks how links differ by sex, so Model 2 adds one interaction, enjoyment × sex. Only one is tested, as 32 rows cannot support more.

```
model_2 <- lm(log_vpr ~ mot_health + mot_enjoy + bar_time + bar_health +
                mot_enjoy:male + group + year, data = gym)
summary(model_2)
```

```
##
## Call:
## lm(formula = log_vpr ~ mot_health + mot_enjoy + bar_time + bar_health +
##     mot_enjoy:male + group + year, data = gym)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.078662 -0.027477  0.001157  0.023836  0.099485
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -0.623118   0.104588  -5.958 2.01e-05 ***
## mot_health         0.051455   0.028192   1.825 0.086708 .
## mot_enjoy          0.547039   0.094789   5.771 2.86e-05 ***
## bar_time          -0.005846   0.049639  -0.118 0.907713
## bar_health         0.214347   0.077790   2.755 0.014075 *
```

```
## group13-19 Male    1.843293   0.165749  11.121 6.14e-09 ***
## group20-39 Female  0.562227   0.135795   4.140 0.000769 ***
## group20-39 Male    1.378119   0.109483  12.587 1.03e-09 ***
## group40-59 Female -0.103081   0.169929  -0.607 0.552616
## group40-59 Male    0.448126   0.117828   3.803 0.001562 **
## group60+ Female   -0.859803   0.203319  -4.229 0.000639 ***
## group60+ Male     -0.002011   0.208479  -0.010 0.992421
## year2023           0.226850   0.046276   4.902 0.000160 ***
## year2024           0.168578   0.079072   2.132 0.048858 *
## year2025           0.268830   0.085221   3.155 0.006137 **
## mot_enjoy:male    -0.273515   0.106118  -2.577 0.020246 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06534 on 16 degrees of freedom
## Multiple R-squared:  0.9977, Adjusted R-squared:  0.9955
## F-statistic: 459.5 on 15 and 16 DF,  p-value: < 2.2e-16
```

4b. Can the models be compared?

Yes. Both use the same outcome and 32 rows, and Model 1 is nested in Model 2, so adjusted R^2, AIC and an F-test apply.

```
bind_rows(glance(model_1) |> mutate(model = "Model 1: control"),
          glance(model_2) |> mutate(model = "Model 2: + enjoyment x sex")) |>
  select(model, r.squared, adj.r.squared, sigma, AIC, df.residual, nobs) |>
  kable(booktabs = TRUE, digits = 3, caption = "Model fit")
```

Table 9: Model fit

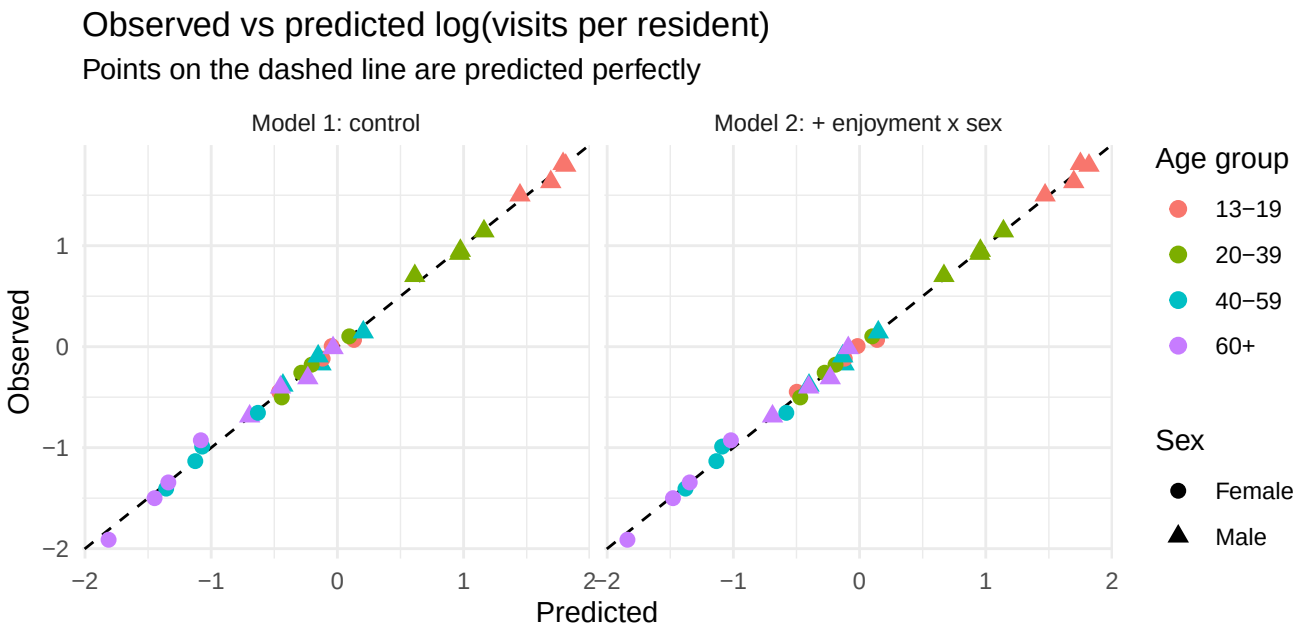
model	r.squared	adj.r.squared	sigma	AIC	df.residual	nobs
Model 1: control	0.997	0.994	0.075	-62.855	17	32
Model 2: + enjoyment x sex	0.998	0.996	0.065	-71.968	16	32

```
f_test <- anova(model_1, model_2)
f_test
```

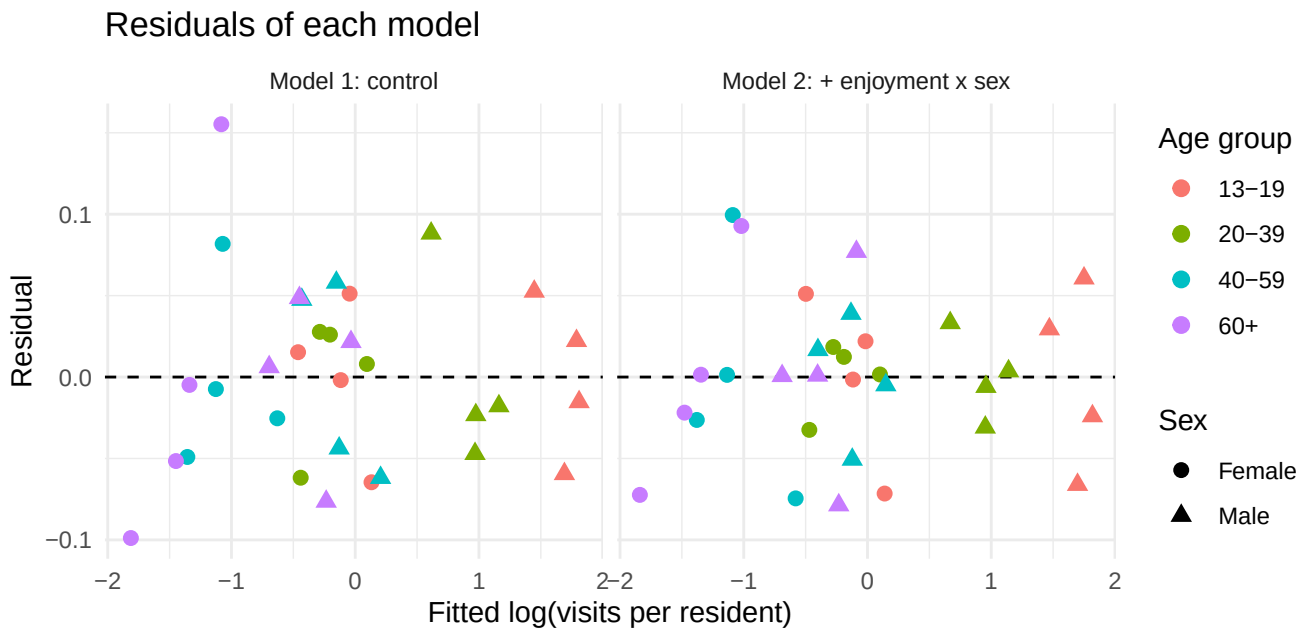
```
## Analysis of Variance Table
##
## Model 1: log_vpr ~ mot_health + mot_enjoy + bar_time + bar_health + group +
##     year
## Model 2: log_vpr ~ mot_health + mot_enjoy + bar_time + bar_health + mot_enjoy:male +
##     group + year
##   Res.Df      RSS Df Sum of Sq      F  Pr(>F)
## 1     17 0.096678
## 2     16 0.068313  1  0.028364 6.6433 0.02025 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4c. Which is the better model?

```
bind_rows(
  gym |> mutate(model = "Model 1: control", predicted = fitted(model_1)),
  gym |> mutate(model = "Model 2: + enjoyment x sex", predicted = fitted(model_2))) |>
  ggplot(aes(predicted, log_vpr, colour = age, shape = sex)) +
  geom_abline(linetype = "dashed") +
  geom_point(size = 2.5) +
  facet_wrap(~ model) +
  labs(title = "Observed vs predicted log(visits per resident)",
       subtitle = "Points on the dashed line are predicted perfectly",
       x = "Predicted", y = "Observed", colour = "Age group", shape = "Sex") +
  theme_minimal()
```




```
bind_rows(
  gym |> mutate(model = "Model 1: control",
                fitted = fitted(model_1), residual = resid(model_1)),
  gym |> mutate(model = "Model 2: + enjoyment x sex",
                fitted = fitted(model_2), residual = resid(model_2))) |>
  ggplot(aes(fitted, residual, colour = age, shape = sex)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_point(size = 2.5) +
  facet_wrap(~ model) +
  labs(title = "Residuals of each model",
       x = "Fitted log(visits per resident)", y = "Residual",
       colour = "Age group", shape = "Sex") +
  theme_minimal()
```




```r
# Check 1: do the four indices add to a group + year baseline?
model_0 <- lm(log_vpr ~ group + year, data = gym)
p_base  <- ptxt(anova(model_0, model_1)[["Pr(>F)"]][2])

# Check 2: fill items missing from the top 10 with each group's 10th-ranked value
fill_tenth <- function(raw, item_col, patterns) {
  tenth <- raw |>
    filter(Year %in% YEARS) |>
    mutate(AgeGroup = clean_age(AgeGroup)) |>
    group_by(Year, AgeGroup, Sex) |>
    summarise(tenth = min(Percentage), .groups = "drop")
  extract_items(raw, item_col, patterns) |>
    left_join(tenth, by = c("Year", "AgeGroup", "Sex")) |>
    mutate(across(all_of(names(patterns)), ~ coalesce(.x, tenth))) |>
    select(-tenth)
}

survey_sens <- fill_tenth(mot_raw, "Motivations", mot_items) |>
  left_join(fill_tenth(bar_raw, "Barriers", bar_items), by = c("Year", "AgeGroup", "Sex")) |>
  transmute(year = factor(Year), age = AgeGroup, sex = Sex,
            mot_health = (physical + mental) / 2,
            mot_enjoy  = (interest + social) / 2,
            bar_time   = overtime + family + school + studywork + twojobs,
            bar_health = too_old + health_cnd) |>
  mutate(across(mot_health:bar_health, ~ as.numeric(scale(.x))))

gym_sens <- gym |>
  select(-(mot_health:bar_health)) |>
  mutate(age = as.character(age), sex = as.character(sex)) |>
  left_join(survey_sens, by = c("year", "age", "sex"))

model_1_sens <- update(model_1, data = gym_sens)
model_2_sens <- update(model_2, data = gym_sens)
```

Model 2 is better: adjusted R^2 rises from 0.994 to 0.996, AIC falls from -62.9 to -72, the F-test is significant (p =

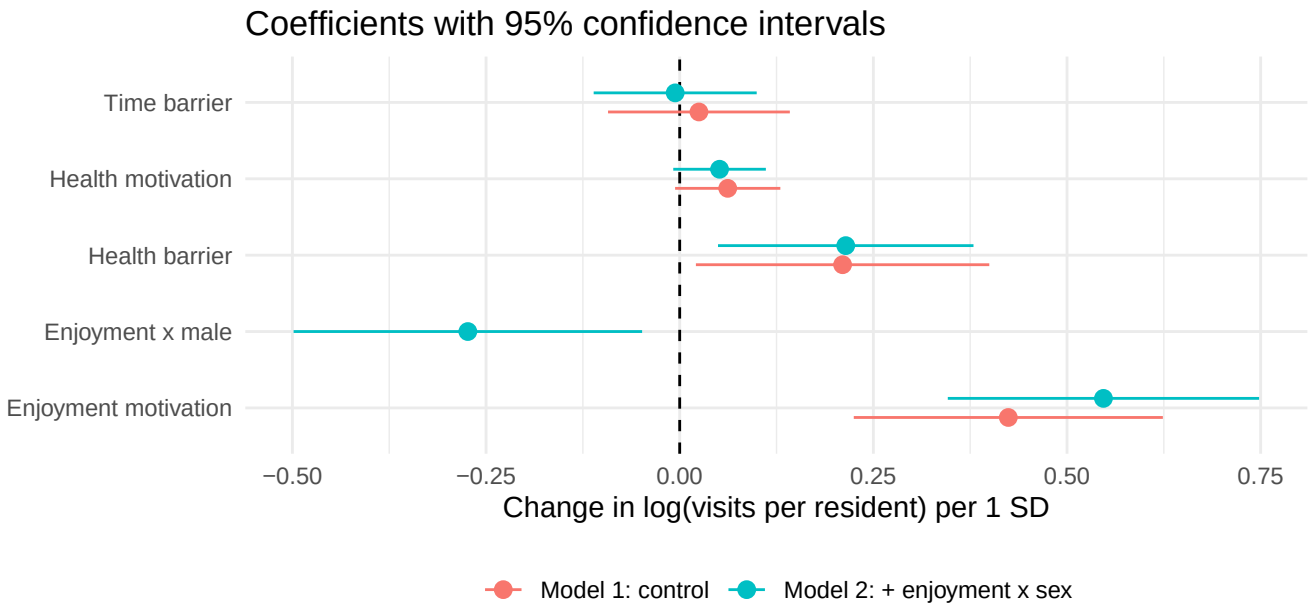
0.020), and its residuals are smaller. However, the interaction was chosen after exploring the data, so the result is tentative.

Two checks support the indices: adding them to a group-and-year baseline improves fit ($p = 0.004$), and filling missing top-10 items with each group's 10th-ranked value (their maximum) instead of 0 barely changes the results (enjoyment +58%, $p < 0.001$; interaction $p = 0.011$).

4d. Interpreting the results

```
coefs <- bind_rows(tidy(model_1, conf.int = TRUE) |> mutate(model = "Model 1: control"),
                   tidy(model_2, conf.int = TRUE) |> mutate(model = "Model 2: + enjoyment x sex")) |>
  filter(term %in% c("mot_health", "mot_enjoy", "bar_time", "bar_health", "mot_enjoy:male")) |>
  mutate(term = recode(term, mot_health = "Health motivation", mot_enjoy = "Enjoyment motivation",
                       bar_time = "Time barrier", bar_health = "Health barrier",
                       `mot_enjoy:male` = "Enjoyment x male"))

ggplot(coefs, aes(estimate, term, colour = model)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_pointrange(aes(xmin = conf.low, xmax = conf.high),
                  position = position_dodge(width = 0.5)) +
  labs(title = "Coefficients with 95% confidence intervals",
       x = "Change in log(visits per resident) per 1 SD", y = NULL, colour = NULL) +
  theme_minimal() + theme(legend.position = "bottom")
```



- **Model 1.** Comparing each group with itself, enjoyment motivation stands out: in years when a group's enjoyment index was 1 SD higher, its visit rate was about 53% higher ($p < 0.001$). Health motivation shows at most a weak link ($p = 0.071$) and time barriers none ($p = 0.661$). The health barrier is positive (23%, $p = 0.032$), which is counter-intuitive; as barriers come from non-participants, we do not read this as an effect.
- **Model 2.** The enjoyment link may be stronger for women (about 73% per 1 SD, $p < 0.001$) than for men (about 31%, $p = 0.015$); the difference is significant ($p = 0.020$) but exploratory.
- **In plain English.** Gym use per resident was higher in years when more people in a group said they exercise for enjoyment and to meet others, possibly more so among women.

Limitations

Results are group-level associations, not individual or causal effects. Motivations and barriers come from different respondents, visits include repeat visits, and the surveys cover sport generally, not only gyms.

References

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